

University Important Questions

UNIT-1

2 Marks

1. What is an algorithm? (UQ April 2012 & APRIL 2013 Nov' 2016) (Ref.Pg.No.1 Qn.No.1)
2. What is best-case efficiency? (UQ APRIL 2013) (Ref.Pg.No.3 Qn.No.18)
3. What is binary search? (UQ April'12) (Ref.Pg.No.7 Qn.No.40)
4. Discuss about various asymptotic notation (UQ Nov'12) (Ref.Pg.No.8 Qn.No.45)
5. What is recursive algorithm? (UQ Nov'12, Apr/May'14) (Ref.Pg.No.8 Qn.No.46)
6. Define sorting (UQ Nov'12, Apr/May'14) (Ref.Pg.No.8 Qn.No.47)

11 Marks

1. Explain about analysis of algorithms & analysis of control structures. (UQ APRIL'13) (Ref.Pg.No.9 Qn.No.1)
2. Write about solving recurrences or Discuss about recursive and non recursive algorithms.(UQ APRIL12) (UQ:Apr/May'14 /May'16) (Ref.Pg.No.15 Qn.No.2)
3. Write and analyze about heap sort (UQ Nov'12) (Ref.Pg.No.30 Qn.No.3)
4. Write in detail about radix sort (UQ APRIL12) (Ref.Pg.No.33 Qn.No.4)
5. Write and analyze selection sort(UQ APRIL13 & Nov'12/Nov'16) (Ref.Pg.No.35 Qn.No.6)
6. Write and analyze about bubble sort (UQ APRIL 13) (Ref.Pg.No.37 Qn.No.7)
7. Write in detail about searching or Explain about Sequential and Fibonacci search (UQ APRIL13 & Nov'10) (UQ: Apr/May'14) (Ref.Pg.No.38 Qn.No.8)
8. Write and analyze Insertion sort.(May'16)

UNIT-2

2 Marks

1. Define the divide and conquer method. (UQ APRIL'13 & APRIL'12) (Ref.Pg.No.1 Qn.No.1)
2. What is the binary search? (UQ APRIL'12) (Ref.Pg.No.1 Qn.No.5)
3. What is the Quick sort? (UQ Nov'10& APRIL'12) (Ref.Pg.No.2 Qn.No.8)
4. Is insertion sort better than the merge sort? (UQ APRIL'13) (Ref.Pg.No.2 Qn.No.10)
5. What is Knapsack problem? (UQ Nov'12) (Ref.Pg.No.3 Qn.No.17)
6. Define minimum spanning tree (UQ Nov'12) (Ref.Pg.No.4 Qn.No.26)
7. What is greedy method? (UQ:Apr/May'14) (Ref.Pg.No.3 Qn.No.19)

11 Marks

1. Write algorithm and explain binary search (UQ APRIL'13 & APRIL'12/May'16)(Ref.Pg.No.6 Qn.No.2)
2. Explain Maximum and Minimum (UQ APRIL'13/ Nov'16) (Ref.Pg.No.8 Qn.No.3)
3. Explain merge sort (UQ APRIL'12,APRIL/MAY'14) (Ref.Pg.No.10 Qn.No.4)
4. Explain strasson's matrix multiplication (UQ Nov'12& APRIL'12) (Ref.Pg.No.15 Qn.No.6)
5. Explain minimum spanning tree (UQ Nov'12,Apr/May'14/Nov'16/ May'14) (Ref.Pg.No.21 Qn.No.10)
6. Explain optimal storage on tapes (UQ APRIL'12) (Ref.Pg.No.30 Qn.No.12)
7. Explain quick sort (UQ:APRIL/MAY'14) (Ref.Pg.No.13 Qn.No.5)

UNIT-3

2 Marks

- 1 Define dynamic programming. (UQ APRIL'12) (Ref.Pg.No.1 Qn.No.2)
2. Write the general procedure of dynamic programming. (UQ APRIL'13) (Ref.Pg.No.1 Qn.No.5)
3. List different traversal technique in graph (UQ APRIL'12) (Ref.Pg.No.77 Qn.No.2)
4. What is biconnected component? (UQ Nov'12) (Ref.Pg.No.2 Qn.No.15)
5. Write about multistage graph with example (UQ APRIL'13 & Nov'12, Apr/May'14) (Ref.Pg.No.2 Qn.No.17)
6. What is all pair shortest path problem?(APRIL/MAY'14/May'16) (Ref.Pg.No.2 Qn.No.18)

11 Marks

1. Explain Travelling salesman problem (UQ APRIL'13 & NOV'12) (Ref.Pg.No.13 Qn.No.4)
2. Explain chained matrix multiplication (UQ NOV'12 & NOV'10/May'16) (Ref.Pg.No.18 Qn.No.5)
3. Explain 0/1 knapsack problem (UQ APRIL'12) (Ref.Pg.No.21 Qn.No.6)
4. Explain basic search and traversal technique (UQ APRIL'13) (Ref.Pg.No.24 Qn.No.8)
5. Explain topological sorting (UQ NOV'10, APRIL/MAY'14) (Ref.Pg.No. 30 Qn.No.10)
6. Explain multistage graph with example (UQ :APRIL/MAY'14/Nov'16) (Ref.Pg.No.3,Qn.No.2)
- 7.Explain Biconnected components (UQ:APRIL/MAY'14)) (Ref.Pg.No.26,Qn.No.9)
- 8.What are AND/OR graph?Explain in detail.(Nov'16)

UNIT-4

2 Marks

- 1 What are the requirements that are needed for performing Backtracking? (UQ NOV'10) (Ref.Pg.No.1 Qn.No.1)
2. State Sum of Subsets problem. (UQ APRIL'13 & APRIL'12) (Ref.Pg.No.2 Qn.No.14)
3. State m – colorability decision problem (UQ NOV'10) (Ref.Pg.No.2 Qn.No.15)
4. Define chromatic number of the graph. (UQ APRIL'13) (Ref.Pg.No.2 Qn.No.16)
5. Explain Hamiltonian cycles? (UQ NOV'12 & APRIL'12,APRIL/MAY'14) (Ref.Pg.No.2 Qn.No.18)
6. Define Backtracking (UQ APRIL'12,APRIL/MAY'14) (Ref.Pg.No.2 Qn.No.19)

11 Marks

1. Explain general Backtracking method (UQ NOV'12) (Ref.Pg.No.3 Qn.No.1)
2. Explain 8-Queens Problem? (UQ APRIL'13 & APRIL '12,APRIL/MAY'14/May/Nov'16) (Ref.Pg.No.4 Qn.No.2)
3. Explain Hamiltonian cycles? (UQ NOV'10/ Nov'16 (Ref.Pg.No.8 Qn.No.4)
4. Explain graph coloring (UQ APRIL'12,APRIL/MAY'14) (Ref.Pg.No.9 Qn.No.6)
5. Explain Knapsack Problem using Backtracking (UQ APRIL'13 & NOV'12) (Ref.Pg.No.12 Qn.No.7)
- 6.Explain sum of Subsets problem and give the algorithm to find the same(May'16)

UNIT-5

2 Marks

- 1 Define Branch-and-Bound method. (UQ APRIL'13 & APRIL'12) (Ref.Pg.No.1 Qn.No.1)
2. What are NP- hard and Np-complete problems? (UQ NOV'12) (Ref.Pg.No.1 Qn.No.3)
3. Explain Least Cost (LC) Search (UQ APRIL'13) (Ref.Pg.No.1 Qn.No.7)
- 4 .FIFO Branch and Bound (UQ NOV'12) (Ref.Pg.No. Qn.No.8)
5. Differentiate backtracking and branch-and-bound. Techniques (UQ APR'11) (Ref.Pg.No.1 Qn.No.9)
6. Give the mathematical equation for 0/1knapsack problem. (UQ APR'11) (Ref.Pg.No.2 Qn.No.18)
7. What is bounding function? (UQ:APRIL/MAY'14) (Ref.Pg.No.3 Qn.No.19)
8. what is non deterministic algorithm (UQ:APRIL/MAY'14)) (Ref.Pg.No.3 Qn.No.20)

11 Marks

1. Explain Branch and bound method (UQ APRIL'12 & APRIL'13,APRIL/MAY'14 Nov'16) (Ref.Pg.No.4 Qn.No.1)
2. Explain Least Cost (LC) Search with 15 puzzle example(UQ APRIL'13) (Ref.Pg.No.6 Qn.No.2) 3. Explain Knapsack problem (UQ NOV'10) (Ref.Pg.No.13 Qn.No.4)
4. Explain travelling salesman problem (UQ NOV'12, NOV'10 & APRIL'12,May/Nov'16) (Ref.Pg.No.21 Qn.No.5)
5. Explain NP hard and NP-completeness (UQ APRIL'13, NOV'12 & APRIL'12 ,APRIL/MAY'14/16) (Ref.Pg.No.29 Qn.No.6)